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HEAT PIPE

Abstract

A heat pipe includes a pipe body having an evaporation section, a condensation section, and a transmission section, the evaporation section and the condensation section are respectively connected to opposite ends of the transmission section, and a composite capillary structure including a first capillary structure and at least one second capillary structure, the first capillary structure having a grooved shape and being disposed on an inner surface of the pipe body, the second capillary structure being at least partially disposed on the first capillary structure, the first capillary structure and the at least one second capillary structure extending from the evaporation section to the condensation section, wherein the composite capillary structure has a variable thickness within the evaporation section.

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Background/Summary

RELATED APPLICATIONS

[0001] This US application claims the benefit of priority to China application No. 202410240817.X, filed on Mar. 1, 2024, of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present utility model relates to a heat pipe, particularly for heat pipe with a capillary structure having a variable thickness.

BACKGROUND

[0003] Electronic devices or machines generate heat with a high temperature during operation, and manufacturers often install heat pipes to dissipate the heat. The heat pipes achieve rapid temperature uniformity by utilizing the internal cooling fluid's evaporation and condensation.

[0004] In detail, after the liquid cooling fluid in the heat pipe absorbs heat at the evaporation end, the liquid cooling fluid vaporizes and moves to the condensation end by vapor pressure. After the gaseous cooling fluid at the condensation end is exothermic and condensed into liquid cooling fluid, the liquid cooling fluid flows back to the evaporation end through capillary structures inside the heat pipe, and then absorbs heat again and evaporates, so as to carry out the cooling cycle. However, the cooling cycle of the cooling fluid through the capillary structure within conventional heat pipes remains inefficient. Consequently, one of the issues that R&D personnel should address is how to enhance the heat pipe's heat dissipation efficacy for electronic devices or machinery during operation.

SUMMARY

[0005] Aspects of the disclosure provide a heat pipe including a pipe body having an evaporation section, a condensation section, and a transmission section, the evaporation section and the condensation section are respectively connected to opposite ends of the transmission section, and a composite capillary structure including a first capillary structure and at least one second capillary structure, the first capillary structure having a grooved shape and being disposed on an inner surface of the pipe body, the second capillary structure being at least partially disposed on the first capillary structure, the first capillary structure and the second capillary structure extending from the evaporation section to the condensation section, wherein the composite capillary structure has a variable thickness within the evaporation section.

[0006] In an embodiment, a thickness of the first capillary structure within the evaporation section can be gradually increased from one end away from the condensation section to one end close to the condensation section, so that the variable thickness of the composite capillary structure varies within the evaporation section.

[0007] In an embodiment, the first capillary structure within the evaporation section can have a length greater than or equal to 10 mm and smaller than or equal to 200 mm.

[0008] In an embodiment, the thickness of the first capillary structure within the evaporation section and at one end close to the condensation section can be greater than or equal to 0.1 mm and smaller than or equal to 1 mm.

[0009] In an embodiment, the thickness of the first capillary structure within the evaporation section and at one end close to the condensation section can be greater than or equal to 0.1 mm and smaller than or equal to 0.5 mm.

[0010] In an embodiment, the first capillary structure within the evaporation section can have a slope surface, where an angle between the slope surface and the inner surface of the pipe body can be greater than or equal to 0.1 degree and smaller than or equal to 6.5 degree.

[0011] In an embodiment, the angle between the slope surface and the inner surface of the pipe body can be greater than or equal to 0.1 degree and smaller than or equal to 3.5 degree.

[0012] In an embodiment, the first capillary structure within the evaporation section can have a plurality of spacers disposed on the inner surface of the pipe body, the at least one second capillary structure within the evaporation section is disposed on the plurality of spacers and the pipe body so that the variable thickness of the composite capillary structure within the evaporation section has intermittent changes.

[0013] In an embodiment, the at least one second capillary structure disposed on the plurality of spacers can have a first thickness and the at least one second capillary structure disposed on the pipe body has a second thickness, where the first thickness is smaller than the second thickness.

[0014] In an embodiment, the first capillary structure can have a thickness that is greater than or equal to 0.01 mm and smaller than or equal to 1 mm.

[0015] In an embodiment, the thickness of the first capillary structure can be greater than or equal to 0.01 mm and smaller than or equal to 0.7 mm.

[0016] In an embodiment, the first thickness or the second thickness of the at least one second capillary structure can be greater than or equal to 0.3 mm and smaller than or equal to 10 mm.

[0017] In an embodiment, a ratio between a total length of the plurality of spacers and a total length of gaps between two adjacent spacers can be great than or equal to 1/10 and smaller than or equal to 1. For example, the ratio between the total length of the plurality of spacers and the total length of gaps between two adjacent spacers can be $\frac{1}{3}$.

[0018] In an embodiment, a ratio between a total length of the plurality of spacers and a total length of gaps between two adjacent spacers can be great than or equal to 1 and smaller than or equal to 10.

[0019] In an embodiment, the at least one second capillary structure can be metal powder, metal braided mesh, or a combination thereof.

[0020] In an embodiment, the at least one second capillary structure within the evaporation section can have a variable thickness.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0021] Aspects of the present disclosure can be understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be increased or reduced for clarity of discussion.

[0022] FIG. 1 illustrates a perspective view of a heat pipe 10 according to aspects of the present disclosure.

[0023] FIG. 2 illustrates a cross-sectional view of the heat pipe 10 as shown in FIG. 1.

[0024] FIG. 3 illustrates a partially enlarged cross-sectional view of the heat pipe 10 as shown in FIG. 1.

[0025] FIG. 4 illustrates a cross-sectional view of the heat pipe 10 along the A-A cutline in FIG. 3.

[0026] FIG. 5 illustrates a cross-sectional view of the heat pipe 10 along the B-B cutline in FIG. 3.

[0027] FIG. 6 illustrates a cross-sectional view of a heat pipe 10A according to aspects of the present disclosure.

[0028] FIG. 7 illustrates a partially enlarged cross-sectional view of the heat pipe 10A as shown in FIG. 6.

[0029] FIG. 8 illustrates a cross-sectional view of the heat pipe 10A along the C-C cutline in FIG. 7.

[0030] FIG. 9 illustrates a cross-sectional view of the heat pipe 10A along the D-D cutline in FIG.

7.

DETAILED DESCRIPTION

[0031] Detailed descriptions and technical contents of the present invention are illustrated below in conjunction with the accompanying drawings. However, it is to be understood that the descriptions and the accompanying drawings disclosed herein are merely illustrative and exemplary and not intended to limit the scope of the present invention.

[0032] Please refer to FIG. 1 to FIG. 3. FIG. 1 illustrates a perspective view of a heat pipe **10** according to aspects of the present disclosure. FIG. 2 illustrates a cross-sectional view of the heat pipe **10** as shown in FIG. 1. FIG. 3 illustrates a partially enlarged cross-sectional view of the heat pipe **10** as shown in FIG. 1.

[0033] The heat pipe **10** can be used to accommodate a cooling fluid (not shown), for example, water, refrigerant, or alcohol such as methanol or ethanol. The heat pipe **10** includes a pipe body **11** and a composite capillary structure **12**. The pipe body **11** include an evaporation section **111**, a condensation section **112**, and a transmission section **113**. The evaporation section **111** and the condensation section **112** are respectively connected to opposite ends of the transmission section **113**. The evaporation section **111** can be used to thermally coupled to a heat source (not shown), such as a central processing unit (CPU) or graphics processing unit (GPU), to absorb the heat generated by the heat source. The “thermally coupled” is referring to either directly thermal contact or connection, or indirectly thermal contact or connection through other thermally conductive media.

[0034] Please also refer to FIG. 4 and FIG. 5. FIG. 4 illustrates a cross-sectional view of the heat pipe **10** along the A-A cutline in FIG. 3. FIG. 5 illustrates a cross-sectional view of the heat pipe **10** along the B-B cutline in FIG. 3.

[0035] The composite capillary structure **12** includes a first capillary structure **121** and a second capillary structure **122**. The first capillary structure **121** has a grooved shape and disposed on an inner surface **114** of the pipe body **11**. The first capillary structure **121** can be intergraded with the pipe body **11**. The second capillary structure **122** can be disposed on the first capillary structure **121**. The second capillary structure **122** can be formed using, for example, a metal powder, a metal braided mesh, a combination thereof, or any other suitable material for the specific application. The first capillary structure **121** and the second capillary structure **122** extend from the evaporation section **111** to the condensation section **112**. The permeability of the first capillary structure **121** is greater than the permeability of the second capillary structure **122**. In addition, the capillary force of the first capillary structure **121** is less than the capillary force of the second capillary structure **122**.

[0036] The first capillary structure **121** within the evaporation section **111** has its thickness gradually increased from one end away from the condensation section **112** towards the other end close to the condensation section **112**. That is, the first capillary structure **121** has a sloped surface **1211** within the evaporation section **111**. The length of the first capillary structure **121** within the evaporation section **111** depends on the size of the heat source. For example, the larger the size of the heat source is, the longer the length of the first capillary structure **121** within the evaporation section **111** will be.

[0037] Because the thickness of the first capillary structure **121** increases gradually within the evaporation section **111**, and the capillary force of the first capillary structure **121** is less than the second capillary structure **122**, the composite capillary structure **12** in the evaporation section **111** has a greater capillary force closer to the heat source. In this way, the thinnest part of the composite capillary structure **12** within the evaporation section **111** can have a great capillary force from the second capillary structure **122**, which allows the cooling fluid to continuously flow back to the evaporation section **111** to delay or even avoid the dry burning situation. On the other side, the thickest part of the composite capillary structure **12** in the evaporation section **111** has a great permeability from the first capillary structure **121** and a great capillary force from the second

capillary structure **122**, which allows the cooling fluid to quickly flow back to the evaporation section **111**. The composite capillary structure **12** having the gradually increased thickness in the evaporation section **111** also reduces the thermal resistance of the evaporation section **111**, which, in turn, can improve the heat dissipation efficiency of the heat pipe **10**.

[0038] The first capillary structure **121** within the evaporation section **111** can have a length **L**. The length **L** may be greater than or equal to 10 mm and smaller than or equal to 200 mm. Depends on the size of the heat source that will be attached to the heat pipe **10**, the length **L** of the first capillary structure **121** within the evaporation section **111** can be adjusted accordingly to suit the specific heat source. A thickness **T1** of one end of the first capillary structure **121** within the evaporation section **111** that is close to the condensation section **121** may be greater than or equal to 0.1 mm and smaller than or equal to 1 mm. More preferably, the thickness **T1** may be greater than or equal to 0.1 mm and smaller than or equal to 0.5 mm. An angle **R** between sloped surface **1211** and the inner surface **114** may be greater than or equal to 0.1 degree and smaller than or equal to 6.5 degree. More preferably, the angle **R** may be greater than or equal to 0.1 degree and smaller than or equal to 3.5 degree. However, the length **L**, thickness **T1**, and angle **R** are not limited to these. The length **L**, thickness **T1**, and angle **R** can have any suitable dimensions to suit the specific application of a heat dissipating device.

[0039] In this embodiment, the first capillary structure **121** is integrated with the pipe body **11**, but is not limited to this. In some embodiments, the first capillary structure **121** can be a separated entity and disposed on the pipe body **11**. For example, the first capillary structure **121** can be soldered onto the pipe body **11**. In this embodiment, the heat pipe **10** only includes one the second capillary structure **122**, but is not limited this. In some embodiments, the number of the second capillary structure can be two or more to further improve the heat dissipation efficiency.

[0040] As shown in FIG. 2, the liquid cooling fluid can vaporize into gaseous cooling fluid after absorbs the heat from the heat source thermally coupled to the evaporation section **111**. The gaseous cooling fluid can flow in the **X** direction through the transmission section **113** to the condensation section **112** and condenses back into the liquid cooling fluid. Due to the permeability of the first capillary structure **121** and the capillary force of the second capillary structure **122**, the liquid cooling fluid can then flow back to the evaporation section **111** in the **Y** direction to achieve the effect of the cooling cycle.

[0041] In the above embodiment, the first capillary structure **121** within the evaporation section **111** is a continuous sloped structure, but is not limited to this. The heat pipe **10A** is an example. FIG. 6 illustrates a cross-sectional view of a heat pipe **10A** according to aspects of the present disclosure, and FIG. 7 illustrates a partially enlarged cross-sectional view of the heat pipe **10A** as shown in FIG. 6.

[0042] The heat pipe **10A** is similar to the heat pipe **10** as shown in FIG. 1, a cooling fluid (not shown), for example, water, refrigerant, or alcohol such as methanol or ethanol can be used within the heat pipe **10A**. The heat pipe **10A** can include a pipe body **11** and a composite capillary structure **12A**. The pipe body **11** can include an evaporation section **111**, a condensation section **112**, and a transmission section **113**. The evaporation section **111** and the condensation section **112** are respectively connected to opposite ends of the transmission section **113**. The evaporation section **111** can be used to thermally coupled to a heat source (not shown), such as a central processing unit (CPU) or graphics processing unit (GPU), to absorb the heat generated by the heat source.

[0043] Please also refer to FIG. 8 and FIG. 9. FIG. 8 illustrates a cross-sectional view of the heat pipe **10A** along the C-C cutline in FIG. 7. FIG. 9 illustrates a cross-sectional view of the heat pipe **10A** along the D-D cutline in FIG. 7.

[0044] The composite capillary structure **12A** includes a first capillary structure **121A** and a second capillary structure **122A**. The first capillary structure **121A** has a grooved shape and disposed on the pipe body **11**. The first capillary structure **121A** can be intergraded with the pipe body **11**. The

second capillary structure **122A** can be disposed on the first capillary structure **121A**. The second capillary structure **122A** can be formed using, for example, a metal powder, a metal braided mesh, a combination thereof, or any other suitable material for the specific application. The first capillary structure **121A** and the second capillary structure **122** extend from the evaporation section **111** to the condensation section **112**.

[0045] The first capillary structure **121A** within the evaporation section **111** includes multiple spacers **121A1** that are disposed on the pipe body **11**. That is, the first capillary structure **121A** within the evaporation section **111** has a non-continuous spacer structure. The second capillary structure **122A** within the evaporation section **111** is disposed on the pipe body **11** and on these spacers **121A1**. The thickness of the second capillary structure **122A** disposed on the spacer **121A1** is smaller than the thickness of the second capillary structure **122A** disposed on the pipe body **11**. In this way, the thickness of the composite capillary structure **12A** within the evaporation section **111** has intermittent change. In other word, the thinnest portion of the composite capillary structure **12A** is where the second capillary structure **122A** is disposed directly on the pipe body, and the thickest portion of the composite capillary structure **12A** is where the second capillary structure **122A** is disposed on the spacer **121A1** of the first capillary structure **121A**. In the evaporation section **111**, the second capillary structure **122A** disposed on the pipe body **11** is connected with the second capillary structure **122A** disposed on the spacer **121A1**. The length of the first capillary structure **121A** within the evaporation section **111** depends on the size of the heat source. For example, the larger the size of the heat source is, the longer the length of the first capillary structure **121A** within the evaporation section **111** will be.

[0046] The evaporation section **111** of the heat pipe **10A** can have a great capillary force provided by the second capillary structure **122A** to delay or even avoid the dry burning situation, similar to the heat pipe **10**. In addition, because portions of the second capillary structure **122A** are disposed directly on the pipe body **12A** in the gap between the spacers **121A1**, the thermal resistance can be reduced within the evaporation section **111**. At the portion where the spacers **121A** is located, the composite capillary structure **12** in the evaporation section **111** can have a great permeability from the first capillary structure **121A** and a great capillary force from the second capillary structure **122A**, which can allow the cooling fluid to quickly flow back to the evaporation section **111**. In that way, it improves the heat dissipation efficiency of the heat pipe **10A**.

[0047] The first capillary structure **121A** within the evaporation section **111** has a thickness **T2** that can be greater than or equal to 0.01 mm and smaller than or equal to 1 mm. More preferably, the thickness **T1** can be greater than or equal to 0.01 mm and smaller than or equal to 0.7 mm. The first capillary structure **121A** within the transmission section **113** has a thickness that can be greater than or equal to 0.03 mm and smaller than or equal to 1 mm. The first capillary structure **121A** within the transmission section **113** has a width that can be greater than or equal to 0.03 mm and smaller or equal to 0.6 mm.

[0048] The second capillary structure **122A** within the evaporation section **111** can have a thickness **T3** that can be greater than or equal to 0.3 mm and smaller than or equal to 10 mm, but does not limited to this. The second capillary structure **122A** can have any suitable thickness as long as the thickness of the second capillary structure **122A** disposed on the spacer **121A1** is the same as the thickness of the second capillary structure **122A** disposed on the pipe body **12** between two neighboring spacer **121A1**. The second capillary structure **122A** can have a thickness **T4** within the condensation section **112** and a thickness **T5** within the transmission section. The thickness **T4** and thickness **T5** can be greater than or equal to 0.3 mm and smaller than or equal to 12 mm.

[0049] A ratio between the total length of the spacers **121A1** of the first capillary structure **121A** and the total length of the gaps between two neighboring spacers **121A1** is greater than or equal to 1/10 and smaller than or equal to 1. For example, the ratio between the total length of the spacers **121A1** of the first capillary structure **121A** and the total length of the gaps between two neighboring spacers **121A1** is $\frac{1}{3}$. Therefore, in this embodiment, the total length of the spacers

121A1 can be smaller or equal to the total length of all the gap between two neighboring spacers **121A1**. However, it does not limited to this. In some embodiments, the ratio between the total length of the spacers **121A1** of the first capillary structure **121A** and total length of the gaps between two neighboring spacers **121A1** is greater than or equal to 1 and smaller than or equal to 10. That is, the total length of the spacers **121A1** can be larger or equal to the total length of all the gap between two neighboring spacers **121A1**.

[0050] According to the heat pipe in the abovementioned embodiments, due to the change in the total thickness of the composite capillary structure located in the evaporation section, the composite capillary structure can have great capillary force at its thinnest part provided by the second capillary structure, which can allow the cooling fluid to continuously flow back to the evaporation section to delay or even avoid the dry burning situation. In addition, the thickest part of the composite capillary structure in the evaporation section can have a great permeability from the first capillary structure and a great capillary force from the second capillary structure, which can allow the cooling fluid to quickly flow back to the evaporation section and reduce the thermal resistance of the evaporation section. In this way, it improves the heat dissipation efficiency of the heat pipe.

[0051] Therefore, embodiments disclosed herein are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the embodiments disclosed may be modified and practiced in different but equivalent manners apparent to those of ordinary skill in the relevant art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above may be altered, combined, or modified and all such variations are considered within the scope and spirit of the present disclosure.

[0052] The embodiments illustratively disclosed herein suitably may be practiced in the absence of any element that is not specifically disclosed herein and/or any optional element disclosed herein. While compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods can also “consist essentially of” or “consist of” the various components and steps. All numbers and ranges disclosed above may vary by some number. Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range is specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces.

Claims

1. A heat pipe, comprising: a pipe body having an evaporation section, a condensation section, and a transmission section, the evaporation section and the condensation section are respectively connected to opposite ends of the transmission section; and a composite capillary structure including a first capillary structure and at least one second capillary structure, the first capillary structure having a grooved shape and being disposed on an inner surface of the pipe body, the one second capillary structure being at least partially disposed on the first capillary structure, the first capillary structure and the second capillary structure extending from the evaporation section to the condensation section, wherein the composite capillary structure has a variable thickness within the evaporation section.

2. The heat pipe of claim 1, wherein a thickness of the first capillary structure within the evaporation section is gradually increased from one end away from the condensation section to one

end close to the condensation section, so that the variable thickness of the composite capillary structure varies within the evaporation section.

3. The heat pipe of claim 2, wherein the first capillary structure within the evaporation section has a length greater than or equal to 10 mm and smaller than or equal to 200 mm.
 4. The heat pipe of claim 2, wherein the thickness of the first capillary structure within the evaporation section and at one end close to the condensation section is greater than or equal to 0.1 mm and smaller than or equal to 1 mm.
 5. The heat pipe of claim 4, wherein the thickness of the first capillary structure within the evaporation section and at one end close to the condensation section is greater than or equal to 0.1 mm and smaller than or equal to 0.5 mm.
 6. The heat pipe of claim 2, wherein the first capillary structure within the evaporation section has a slope surface, where an angle between the slope surface and the inner surface of the pipe body is greater than or equal to 0.1 degree and smaller than or equal to 6.5 degree.
 7. The heat pipe of claim 6, wherein the angle between the slope surface and the inner surface of the pipe body is greater than or equal to 0.1 degree and smaller than or equal to 3.5 degree.
 8. The heat pipe of claim 1, wherein the first capillary structure within the evaporation section has a plurality of spacers disposed on the inner surface of the pipe body, the at least one second capillary structure within the evaporation section is disposed on the plurality of spacers and the pipe body so that the variable thickness of the composite capillary structure within the evaporation section has intermittent changes.
 9. The heat pipe of claim 8, wherein the at least one second capillary structure disposed on the plurality of spacers has a first thickness and the at least one second capillary structure disposed on the pipe body has a second thickness, where the first thickness is smaller than the second thickness.
 10. The heat pipe of claim 9, wherein the first capillary structure has a thickness that is greater than or equal to 0.01 mm and smaller than or equal to 1 mm.
 11. The heat pipe of claim 10, wherein the thickness of the first capillary structure is greater than or equal to 0.01 mm and smaller than or equal to 0.7 mm.
 12. The heat pipe of claim 9, wherein the first thickness or the second thickness of the at least one second capillary structure is greater than or equal to 0.3 mm and smaller than or equal to 10 mm.
 13. The heat pipe of claim 9, wherein a ratio between a total length of the plurality of spacers and a total length of gaps between two adjacent spacers is great than or equal to 1/10 and smaller than or equal to 1.
 14. The heat pipe of claim 13, wherein the ratio between the total length of the plurality of spacers and the total length of gaps between two adjacent spacers is $\frac{1}{3}$.
 15. The heat pipe of claim 9, wherein a ratio between a total length of the plurality of spacers and a total length of gaps between two adjacent spacers is great than or equal to 1 and smaller than or equal to 10.
 16. The heat pipe of claim 1, wherein the at least one second capillary structure is metal powder, metal braided mesh, or a combination thereof.
 17. The heat pipe of claim 1, wherein the at least one second capillary structure within the evaporation section has a variable thickness.
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